

Problem 1 — Scaling head-to-head

You have `Weight_kg` = [58, 62, 65, 66, 190] and `Height_cm` = [150, 160, 170, 175, 180].

Tasks:

- Compute the Standardization for both features by hand on the first three values only.
- Compute Min–Max Scaling to [0, 1] for all values.
- Compute Robust Scaling for `Weight_kg` using median and IQR.
- Which method handles the outlier in `Weight_kg` best and why?

Problem 2 — When to use which scaler

You are preparing features for a simple model:

- `Monthly_Income` in BDT with a few extremely high earners
- `Room_Temperature_C` within 18 to 30
- `Transaction_Count_Last_7_Days` with many zeros and a few heavy shoppers

Tasks:

- For each feature, pick one scaler: Standardization, Min–Max, or Robust. Justify each choice in one sentence.
- Give a quick numeric example showing why your chosen scaler is safer or more interpretable for that feature.

Problem 3 — Nominal vs Ordinal and encoding

Dataset:

ID	City	Education_Level	Satisfaction
1	Dhaka	High School	Low
2	Chattogram	Bachelor	Medium
3	Rajshahi	Master	High
4	Dhaka	Bachelor	Medium

Tasks:

- a) Mark which are nominal and which are ordinal.
- b) One-hot encode City by hand for the 4 rows. Use 0 and 1.
- c) Ordinal-encode Education_Level with High School=0, Bachelor=1, Master=2.

Problem 4 — Vectors, dot product, norms

Given vectors $\mathbf{a} = [3, -1, 2]$ and $\mathbf{b} = [4, 0, -2]$.

Tasks:

- a) Compute the dot product $\mathbf{a} \cdot \mathbf{b}$ by hand.
- b) Compute L1 and L2 norms of $\mathbf{a}$.
- c) Write a tiny numpy snippet to verify.

Problem 5 — Euclidean vs Manhattan distance in practice

Points: P1(2, 3), P2(5, 7), P3(2, 10).

Tasks:

- a) Compute Euclidean and Manhattan distances for all pairs by hand.
- b) Which metric changes more when a single coordinate has a large jump, and why?
- c) Scale the y-coordinate by a factor of 10 and recompute distances between P1 and P2. What did this do to your intuition?